

JÚLIO CÉSAR ASSUNÇÃO DE FRANÇA FILHO

Junior Data Scientist | Python | SQL | Machine Learning | Generative AI | RAG

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OBJECTIVE

Junior Data Scientist in data, AI/ML or analytics teams, applying Python, SQL, statistics, machine learning and Generative AI/RAG to turn business problems into metrics, models and recommendations.

PROFESSIONAL SUMMARY

Junior Data Scientist with a degree in Data Science from Fatec Cotia and experience in data analysis, data integration, KPI development, dashboards and analytical automation in an operational environment. Hands-on experience with Python, SQL, Power BI, Power Query and VBA, with applied projects in machine learning, default risk, Generative AI, RAG, LLMs, embeddings and knowledge graphs.

TECHNICAL SKILLS

- **Programming and data:** Python, Pandas, NumPy, SQL, Cypher, Plotly, Git/GitHub.
- **Machine Learning and statistics:** EDA, feature engineering, classification, model evaluation, PR-AUC, ROC-AUC, precision, recall, F1, hypothesis testing and inference.
- **Generative AI and NLP:** LLMs, RAG, embeddings, LangChain, Ollama, prompt engineering, answer evaluation and semantic retrieval.
- **BI, data and development:** Power BI, DAX, Power Query, Advanced Excel, VBA, ETL/ELT, Neo4j, Flask, Streamlit, HTML, CSS, JavaScript.

PROFESSIONAL EXPERIENCE

Data Analysis Intern, Ombudsman Office | Elo Participações Ltda

Feb/2023 - Feb/2025

- Built indicators and dashboards to monitor case deadline extension rates in the Ombudsman Office, supporting a task force that reduced the monthly extension rate from approximately 20%-30% to the company's 10% target.
- Cleaned, standardized and integrated complaint, call and satisfaction survey datasets for operational and management analyses.
- Developed Power BI and Excel dashboards and reports using Power Query, DAX, descriptive statistics and operational KPIs.
- Automated recurring Excel routines with Power Query, VBA and macros, reducing manual effort in data consolidation and updates.
- Supported executive presentations with KPI analysis, trends and key points for decision-making.

RELEVANT PROJECTS

Default Risk Prioritization | Python, Machine Learning, Streamlit

Portfolio

- Developed an end-to-end solution to estimate 5+ day late-payment risk and prioritize collection/risk review queues with human oversight.
- Built a pipeline with temporal validation, feature engineering, supervised training, business baseline, initial calibration, segment analysis, model card and Streamlit app.
- Official result: Random Forest with PR-AUC 0.5546, ROC-AUC 0.9217 and 2.22x PR-AUC lift vs baseline; threshold 0.25 with precision 0.5467, recall 0.6455 and F1 0.5920.

AgroTech - Agribusiness Financial Statement Analysis with Generative AI/RAG

PwC/Fatec Challenge

- Developed, as part of a team, a web application for comparative analysis of agribusiness income statements using Generative AI, RAG, structured data, traceable sources and evaluation metrics.
- Implemented income statement ingestion, deterministic calculations in Python, retrieval with PostgreSQL/pgvector, bge-m3 embeddings, Flask and evaluation on a curated dataset of 60 questions.
- Evaluation metrics: numeric accuracy 1.0000, calculation accuracy 1.0000, source accuracy 1.0000, context precision 0.8619, section completeness 0.9310 and refusal accuracy 1.0000.

AI Chatbot for Arbovirus Surveillance and Control | Capstone in partnership with FAPESP

Aug/2024 - Dec/2025

- Developed an AI prototype to query operational, epidemiological, environmental and official guideline data on arboviruses, integrating SisaWeb, SINAN/DataSUS and NASA POWER.
- Modeled a Neo4j knowledge graph and a RAG layer with official documents, chunking, bge-m3 embeddings, Python, LangChain, Ollama, HTML/CSS/JS and Cypher.
- Result: functional prototype with a 32.1M-node and 120.7M-relationship graph, grounded answers and evaluation with 24 questions across graph, RAG and hybrid scenarios.

EDUCATION

- Degree in Data Science - Fatec Cotia | Completed in 2025.
- Technical High School in Systems Development - ETEC Uirapuru / Centro Paula Souza | Completed in 2021.

COURSES AND LANGUAGES

- **Courses:** Machine Learning A-Z (Udemy); Advanced Python Data Analysis with Pandas (Udemy); Excel from Beginner to Advanced (Udemy); Excel Module II (Impacta).
- **Languages:** Portuguese - native; English - TOEIC score 890/990, Upper-Advanced proficiency, between C1 and C2 levels, test taken in March 2025.